

Skills Summary

Proportional Relationships

Use this reference while completing your worksheet or when studying.

Skill 1 — From a table to an equation

Rule: a table shows a proportional relationship when $\frac{y}{x}$ is the *same* for every column. That common value is k , the constant of proportionality, and the equation is $y = kx$.

To find k : pick any column and divide y by x . Then check a second column — if it gives a different value, the relationship is not proportional.

Example: A table pairs $x = 4$ with $y = 10$, and $x = 6$ with $y = 15$. Find k and write the equation.

Step 1. I want the constant of proportionality, and a table gives me matched pairs — so I divide y by x in one column and check it against another.

Step 2. $\frac{15}{6} = 2.5$, and the other column agrees: $\frac{10}{4} = 2.5$. So the equation is $y = 2.5x$.
 $\Rightarrow k = 2.5$

Try it: A table pairs $x = 3$ with $y = 21$, and $x = 5$ with $y = 35$. Find k .

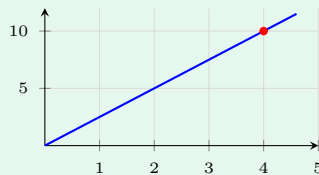
check: $k = 7$

Skill 2 — From a graph to a unit rate

How to spot one: the graph of a proportional relationship is a *straight line through the origin*. A straight line that misses the origin is not proportional.

Unit rate = slope: pick a point (x, y) where the line crosses a corner of the grid and compute $\frac{y - 0}{x - 0}$. It is the value of y when x is 1.

Example: The line below goes through the origin and through $(4, 10)$. Find its unit rate.



Step 1. The line passes through the origin, so the relationship is proportional and its unit rate is just the slope — rise over run from the origin to a lattice point.

Step 2. From $(0, 0)$ to $(4, 10)$ the rise is 10 and the run is 4: $\frac{10}{4} = 2.5$. $\Rightarrow k = 2.5$

Try it: A line through the origin also passes through $(6, 9)$. What is its unit rate?

check: $k = 1.5$

Skill 3 — Using the equation $y = kx$

Given x , want y : substitute and multiply — $y = k \cdot x$.

Given y , want x : that is an equation to *solve*, not a substitution: divide, $x = y \div k$.

Knowing which of the two you are doing is most of the work: read the question for which quantity is already known.

Example: A relationship has $k = 8$. Find y when $x = 7$.

Step 1. The constant and the x -value are both known and y is wanted, so this is a substitution, not an equation to solve.

Step 2. $y = 8 \times 7 = 56$. $\Rightarrow$ **$y = 56$**

Try it: A relationship has $k = 9$. Find y when $x = 6$.

check: $54 = 9 \times 6$

Skill 4 — From a story to a model

Steps: name the two quantities with letters; find one matched pair in the story; divide to get k ; write $y = kx$; then use it.

Careful: a story with a fixed charge *plus* a rate (a joining fee, a monthly line rental) is *not* proportional — doubling the amount does not double the total.

Example: 3 pencils cost 2 dollars. At the same rate, what do 12 pencils cost?

Step 1. The cost rises in step with the number of pencils and one matched pair is given, so I find the cost of a single pencil and multiply.

Step 2. $k = \frac{2}{3}$ dollars a pencil, so 12 pencils cost $\frac{2}{3} \times 12 = 8$ dollars. $\Rightarrow$ **8 dollars**

Try it: 4 pens cost 6 dollars. At the same rate, what do 14 pens cost?

check: $12 = \frac{6}{4} \times 8$